Creating Markets for Carbon Removal

William Arnesen, Rachel Glennerster, Emmanuel Murray-Leclair, Manpreet Singh, Alexander Slaski, Christopher Snyder, and Eric Trueswell
March 2026

Reducing current emissions will not solve the climate crisis alone; removing the carbon accumulated in the atmosphere over a century of emissions will likely also be required. According to the Intergovernmental Panel on Climate Change, even with ambitious emissions abatement, hundreds of billions of tons of carbon dioxide must be removed this century to avoid exceeding a 2.0°C temperature-increase target.[1] However, current technologies for durable carbon removal are too expensive for widespread adoption without further innovation. Methods like direct air capture (DAC), enhanced rock weathering, and ocean alkalinity enhancement are technically feasible, but durable removal remains costly.[2,3,4] For example, according to U.S. Department of Energy estimates, the cost of DAC is currently in the range of \$220–\$600 per ton of CO_2 removed.[5]

Large-scale carbon removal requires a strategy to bring down costs and, once costs have fallen, to integrate durable removal into compliance markets to provide long-term demand. Carbon-pricing systems already mobilize large sums: the World Bank reports carbon-pricing revenues exceeded \$100 billion globally in 2024; the EU Emissions Trading System (ETS) alone generated roughly €38.8 billion in auction revenues that year.[6] Allowing credits for engineered, durable carbon removal to be sold

---

1 Intergovernmental Panel on Climate Change. *Climate Change 2023: Synthesis Report*. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Edited by Hoesung Lee and José Romero. Geneva: IPCC, 2023. doi:10.59327/IPCC/AR6-9789291691647.

2 Yale Center on Natural Carbon Capture. "Enhanced Weathering." Yale University. naturalcarboncapture.yale.edu/research/geological-ocean-capture/enhanced-weathering

3 International Energy Agency. "Direct Air Capture." *IEA Energy System: Carbon Capture, Utilisation and Storage*. www.iea.org/energy-system/carbon-capture-utilisation-and-storage/direct-air-capture

4 Dickhardt, Fabian J., T. Alan Hatton, and Kripa K. Varanasi. "Ocean Alkalinity Enhancement." *MIT Climate Portal*. December 12, 2024. climate.mit.edu/explainers/ocean-alkalinity-enhancement

5 U.S. Department of Energy. *Pathways to Commercial Liftoff: Carbon Management*. April 2023. web.archive.org/web/20240318171107/https://liftoff.energy.gov/wp-content/uploads/2024/02/20230424-Liftoff-Carbon-Management-vPUB_update4.pdf

6 World Bank. "Global Carbon Pricing Mobilizes Over \$100 Billion for Public Budgets." Press release, June 10, 2025. worldbank.org/en/news/press-release/2025/06/10/global-carbon-pricing-mobilizes-over-100-billion-for-public-budgets

in these markets would generate substantial demand for carbon removal. Carbon removal could be an especially promising competitive alternative to emissions abatement in hard-to-abate sectors. Expanding the ETS to cover more sectors at competitive cost levels could help the EU resist pressure to weaken its ambitious net-zero timetable. At present, the EU does not allow removal credits to be traded in the ETS for fear that many are non-additional, actions that would have happened anyway because of associated co-benefits. Durable removal could mitigate these concerns if backed by rigorous measurement, reporting, verification, and additionality standards. Non-additionality is less of a concern for engineered, durable carbon removal in any event, since it lacks associated co-benefits that other offsets, such as reforestation projects, have.[7]

The investment needed to bring down durable carbon-removal costs on the timelines implied by net-zero targets will likely require philanthropic or government subsidies. While emissions costs in the EU ETS are currently well below those of durable carbon removal, ETS prices will rise as hard-to-abate sectors are included and the cheapest emissions-reduction solutions are exhausted. For example, EU carbon permits traded for €81 ($96) per ton of CO_2 in early February 2026.[8] Even so, the financing problem is front-loaded: innovators must fund large R&D and scale-up investment years before there is reliable, policy-backed demand for durable removal. Market forces are unlikely to generate investment at a scale commensurate with the social benefits, even with eventual integration into ETSs. This is because innovators cannot fully capture the returns to this socially valuable information: knowledge spillovers, imitation, and pressure to price this socially valuable commodity closer to marginal cost can prevent innovators from recouping large up-front R&D investments (Arnesen et al., 2026). In other words, compliance-market integration can provide long-run demand, but it does not by itself solve the near-term financing gap before costs fall and policy-backed demand becomes credible.

This financing problem is compounded by uncertainty about which durable carbon-removal technologies will see steep cost declines. Past examples suggest innovation and scaling can sharply reduce the cost of emerging “green” technologies—but not always. Solar-power costs fell 89% between 2009 and 2019[9]

& International Carbon Action Partnership (ICAP). “EU ETS.” Accessed February 2026. icapcarbonaction.com/en/ets/eu-emissions-trading-system-eu-ets

[7] Chen, Qiaoyi, Nicholas Ryan, and Daniel Xu. “Firm Selection and Growth in Carbon Offset Markets: Evidence from the Clean Development Mechanism.” NBER Working Paper 33636, 2025. doi:10.3386/w33636

[8] Trading Economics. “EU Carbon Permits.” Accessed February 2, 2026. tradingeconomics.com/commodity/carbon

[9] Ritchie, Hannah. “Solar Panel Prices Have Fallen by around 20% Every Time Global Capacity Doubled.” *Our World in Data*. June 13, 2024.

after substantial government subsidies, but costs of tidal power have remained high.[10] Investment would ideally be directed toward technologies with plummeting cost curves, but it is hard to predict which these will be in advance. Policies designed to reduce carbon-removal costs should therefore support many approaches early and then scale the most promising.

This brief argues for a three-stage path to scale durable carbon removal. Stage 1 is early portfolio experimentation across technologies. Stage 2 is intermediate, credible commitments to purchase future carbon removal at scale, conditional on costs falling to reasonable levels. Stage 3 is eventual integration into compliance markets to provide long-term demand. In practice, this means moving from smaller Frontier-style portfolio experiments to larger intermediate commitments to purchase future carbon removal at scale, and then to compliance-market integration. The remainder of this brief uses Frontier to illustrate Stage 1 and discusses how best to undertake Stage 2 based on analysis in our companion paper, Arnesen et al. (2026).[11]

Funding for carbon removal is starting to employ elements of Stages 1 and 2. Most investments in carbon removal to date have backed a specific technology (e.g., investment in DAC by the U.S. government).[12] Frontier, a distinctive initiative, is investing corporate funding commitments totaling over $1 billion in a range of carbon-removal technologies, some quite nascent and experimental, in the spirit of Stage 1. Frontier also contains elements of Stage 2: by committing in advance to spend such a considerable sum on carbon removal, Frontier provides an incentive for firms to work on the problem and compete on costs. By making advance purchase agreements with individual carbon-removal firms that include firm-specific cost targets, it incentivizes firms to drive down costs over time.[13]

Substantial Stage 2 funding designed to reduce carbon-removal costs before compliance-market integration will likely need to combine targeted deals with

---

ourworldindata.org/data-insights/solar-panel-prices-have-fallen-by-around-20-every-time-global-capacity-doubled
& Roser, Max. “Why did renewables become so cheap so fast?” Our World in Data. Accessed February 2026. ourworldindata.org/cheap-renewables-growth

[10] Crawford, Iris. “Why Don’t We Use Tidal Power More?” *MIT Climate Portal*. April 12, 2022. climate.mit.edu/ask-mit/why-dont-we-use-tidal-power-more

[11] Arnesen, William, Emmanuel Murray-Leclair, Manpreet Singh, Christopher M. Snyder, Alexander Slaski, and Eric Trueswell. “Designing Mechanisms to Fund Innovation in Greenhouse Gas Removal: Reverse Auctions under Endogenous Entry.” Unpublished CGD working paper, 2026.

[12] U.S. Department of Energy, National Energy Technology Laboratory. “Biden–Harris Administration Announces $3.7 Billion to Kick-Start America’s Carbon Dioxide Removal Industry.” Press release, December 14, 2022. netl.doe.gov/node/12239

[13] Frontier Climate. “An Advance Market Commitment to Accelerate Carbon Removal.” April 12, 2022. frontierclimate.com/writing/launch

specific firms and larger technology-neutral commitments to purchase future carbon removal. Targeted deals are useful for funding startups that find it hard and expensive to raise capital. They are also useful when firms and funders are equally ignorant (or knowledgeable) about how much costs per ton will fall with more investment. But startups are rarely the companies that take innovations to scale. This is likely to require big firms with access to deep (cheaper) pools of capital and the experience of managing large, decades-long engineering projects. This experience means they have better information than funders about whether costs per ton will fall and by how much. Attracting them will require much larger commitments to purchase carbon removal. The fact that they know more than funders means funders need to structure commitments carefully. Specifically, we propose that funding is committed up front but paid only if and when costs have fallen to a reasonable level. This is Stage 2. Under this approach, firms (with their better knowledge), not funders, pick which technology to back.

Where do we look for models of good policy for Stages 1 and 2? For Stage 1, Frontier provides a concrete model of funding experimentation across a technology portfolio. Stage 2 has not been tried at scale, so there are fewer guides for how to design a large-scale committed fund for future low-cost carbon removal. Arnesen et al. (2026) highlight several principles for designing Stage 2, which we summarize below.[11]

Funders should commit now to the auction rules under which they will purchase verified carbon removal in the future. This will spur innovation and scale, generate information on which technologies are cheapest, and fund a lot of carbon removal. The commitment would need to be supported by legally binding contracts and pre-committed funds. The binding nature of the commitment is important because firms will need confidence to make large R&D investments. Under this mechanism, firms take the risk that their technology does not work (as they should), but not the risk that the funder tries to back out of the deal or bargain the price down from the auction outcome. The funder must stick to the pre-specified rules even in the unlikely event that technological advances do not drive carbon-removal costs much below the funder's maximum willingness to pay.

The auction should pay up to the social cost of carbon, if that is what it takes. Procurement auctions are typically designed with features to get as good a deal as possible for the buyer, including capping the maximum bid (called the reserve price) well below the buyer's willingness to pay. But with carbon removal, firms need to be incentivized to make substantial investments in uncertain technologies. In this setting, Arnesen et al. (2026) show that the optimal auction should accept bids up to the social cost of carbon (roughly $250 per ton at current estimates).[14] Ironically, the funder's commitment to give away its full value of carbon removal (if that is how the auction turns out) induces more entry and innovation, resulting in a lower winning bid on average than if the price cap were tighter. The relaxed price cap provides firms with the correct investment incentives without having to resort to a guaranteed price floor.

A coalition of funders making a 10- to 15-year commitment should institutionalize independent measurement, reporting, and verification standards. A technical advisory committee would support these standards and design the future auction rules, determine reserve values, and ensure scientific rigor. Public release of auction data would further enhance accountability and global coordination.[15]

An alternative is to hold an auction now for long-term offtake agreements, locking in prices and quantities with specific firms to deliver carbon removal over the next 10 to 15 years. This is similar to what the UK did for offshore wind. The advantage is that the winning firms can take their contracts to financiers and get a lower cost of capital. The disadvantage is that it risks locking funders into technologies that turn out not to be the best. The analysis in Arnesen et al. (2026) suggests that—when very large sums are being committed—committing to auctions in the future is cheaper despite the higher costs of capital because it spurs competition between firms to find the lowest-cost technology.[16]

In conclusion, integration into compliance markets is likely necessary but not sufficient to fund innovation that will deliver cheaper durable carbon-removal technologies at scale. Achieving cost-competitive carbon removal will require a deliberate sequence: early-stage experimentation across technologies, credible intermediate commitments to purchase removal at scale (conditional on costs falling), and eventual integration into compliance markets for long-term demand. Together, these stages create the demand signal needed to mobilize private investment into a diverse set of technologies and reveal which approaches ultimately compete at scale. Designing markets that do all three is essential if we hope to make net-zero a reality.

---

[14] U.S. Environmental Protection Agency. "Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances." November 2023. epa.gov/system/files/documents/2023-12/epa_scghg_2023_report_final.pdf

[15] Arnesen, William, and Rachel Glennerster. "Market Shaping to Combat Climate Change." In *New Directions in Market Design*, edited by Irene Yuan Lo, Michael Ostrovsky, and Parag A. Pathak. University of Chicago Press, 2024.

[16] On the other hand, early offtake agreements may be better according to Arnesen et al. (2026) if smaller initial quantities are being purchased, if it is clear which technology is likely to emerge as the winner, or if R&D costs are exceptionally high.